



NORTH SOUTH UNIVERSITY

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School of Engineering and Physical Science Department of Mathematics & Physics

Assignment-02

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Course Name and Course Code:

Introduction to Linear Algebra, MAT-125

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The implication of linear equation in real life

Suppose Tomi owns a small business that sells two types of products: Product A, which costs 2 dollars per unit, and Product B, which costs 3 dollars per unit. Tomi sold 36 units of both products in a day combined and earned 32 dollars.

Using this information, we can set up the following system of linear equations:

Equation 1: $2x + y = 32$ (total revenue earned)

Equation 2: $x + 3y = 36$ (total units sold)

In this system, x represents the number of units of Product A sold, and y represents the number of units of Product B sold.

To solve for x and y , we can use the method of substitution:

From Equation 1, we get the following:

$$y = 32 - 2x$$

Substituting this expression for y into Equation 2, we get:

$$x + 3(32 - 2x) = 36$$

Simplifying and solving for x , we get:

$$x = 8$$

Substituting this value of x into Equation 1, we get:

$$2(8) + y = 32$$

Solving for y , we get:

$$y = 16$$

Therefore, the number of units of Product A sold is 8, and the number of units of Product B sold is 16.

(Ans)